THE HILBERT-SAMUEL FUNCTION IN A SHORT EXACT SEQUENCE, AND DEGREES OF THE ASSOCIATED POLYNOMIALS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof ends by leaving the degenerate cases in which one of the modules vanishes to the reader; that sentence is kept verbatim inside the span and the main line is written out.

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Allowed standard prerequisites (editorial).

- Comparison of Hilbert-Samuel functions for a finite-length difference. Invoked in the counted proof at lines 14459-14462. Controls the difference of the functions for two modules differing by a finite length quotient.
- Additivity of the Hilbert-Samuel function in a short exact sequence up to a bounded error. Invoked in the counted proof (lines 14462-14470). Gives the degree estimate required in part (1).

Source reference. Original lemma 00IN: sources/stacks-algebra/pinned-original-source.tex, source lines 12404–12404. Navigation pointer for \ref{lemma-Artin-Rees} used inside the counted span (line 14478); the argument that consumes it is retained in the span.

Source reference. Original lemma 00KB: sources/stacks-algebra/pinned-original-source.tex, source lines 14430–14430. Navigation pointer for \ref{lemma-differ-finite-chi} used inside the counted span (line 14460); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00K6: sources/stacks-algebra/pinned-original-source.tex, source lines 14305–14305. Navigation pointer for \ref{lemma-hilbert-ses} used inside the counted span (line 14459); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a Noetherian local ring. Let $I \subset R$ be an ideal of definition. Let $0 \to M' \to M \to M'' \to 0$ be a short exact sequence of finite R-modules. Then*

1. *if M' does not have finite length, then $\chi_{I,M} - \chi_{I,M''} - \chi_{I,M'}$ is a numerical polynomial of degree $<$ the degree of $\chi_{I,M'}$,*
2. *$\max\{\deg(\chi_{I,M'}), \deg(\chi_{I,M''})\} = \deg(\chi_{I,M})$, and*
3. *$\max\{d(M'), d(M'')\} = d(M)$,*

Proof. We first prove (1). Let $N \subset M'$ be as in Lemma 00K6. By Lemma 00KB the numerical polynomial $\chi_{I,M'} - \chi_{I,N}$ has degree $<$ the common degree of $\chi_{I,M'}$ and $\chi_{I,N}$. By Lemma 00K6 the difference

$$\chi_{I,M}(n) - \chi_{I,M''}(n) - \chi_{I,N}(n-c)$$

is constant for $n \gg 0$. By elementary calculus the difference $\chi_{I,N}(n) - \chi_{I,N}(n-c)$ has degree $<$ the degree of $\chi_{I,N}$ which is bigger than zero (see above). Putting everything together we obtain (1).

Note that the leading coefficients of $\chi_{I,M'}$ and $\chi_{I,M''}$ are nonnegative. Thus the degree of $\chi_{I,M'} + \chi_{I,M''}$ is equal to the maximum of the degrees. Thus if M' does not have finite length, then (2) follows from (1). If M' does have finite length, then $I^nM \to I^nM''$ is an isomorphism for all $n \gg 0$ by Artin-Rees (Lemma 00IN). Thus $M/I^nM \to M''/I^nM''$ is a surjection with kernel M' for $n \gg 0$ and we see that $\chi_{I,M}(n) - \chi_{I,M''}(n) = \text{length}(M')$ for all $n \gg 0$. Thus (2) holds in this case also.

Proof of (3). This follows from (2) except if one of M, M', or M'' is zero. We omit the proof in these special cases. □